

RETZ, Austria

WMO: 110220

Lat: 48.7611N

Lon: 15.9417E

Elev: 1050

StdP: 14.15

Time Zone: 1.00 (EUC)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.8	16.8	5.7	7.7	15.2	10.1	9.6	19.4	28.8	44.7	26.4	42.7	7.7	50	0.536

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.6	87.6	69.5	84.2	68.1	81.1	66.4	71.2	83.6	69.4	81.2	67.7	78.7	7.3	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.9	102.9	77.9	65.2	96.9	75.1	63.7	91.8	73.2	35.7	83.7	34.1	81.4	32.7	78.8	77.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.7	20.4	17.8	DB	9.5	92.8	5.4	3.6	5.6	95.4	2.4	97.5	-0.7	99.6	-4.6	102.2	(4)
(5)			WB	8.4	73.8	5.1	2.1	4.7	75.3	1.7	76.5	-1.2	77.7	-4.9	79.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.6	30.6	34.5	41.8	51.9	59.3	65.7	69.1	68.8	59.8	50.4	41.1	32.9	(6)	
	DBStd	15.39	8.27	8.55	8.12	7.84	6.65	6.68	6.29	6.30	6.45	6.79	7.20	7.47	(7)	
	HDD50	2282	601	437	273	70	7	0	0	0	3	81	280	530	(8)	
	HDD65	5720	1065	855	721	395	199	73	31	33	180	455	718	995	(9)	
	CDD50	2490	1	2	17	128	294	471	593	583	298	92	11	0	(10)	
	CDD65	456	0	0	0	3	22	94	160	151	25	1	0	0	(11)	
	CDH74	3293	0	0	0	18	131	663	1234	1111	132	4	0	0	(12)	
	CDH80	911	0	0	0	1	14	165	378	334	19	0	0	0	(13)	
(14) Wind	WSAvg	8.3	8.4	8.9	9.3	9.1	8.9	7.9	8.4	7.6	8.0	7.5	7.6	7.7	(14)	
(15) Precipitation	PrecAvg	20.7	1.0	0.8	1.3	1.4	2.4	2.8	2.8	2.6	1.9	1.4	1.3	1.1	(15)	
	PrecMax	29.6	2.2	2.2	3.5	3.0	4.2	5.5	6.8	6.1	4.7	3.1	2.5	2.0	(16)	
	PrecMin	15.2	0.2	0.1	0.2	0.0	0.5	0.9	0.7	0.8	0.4	0.2	0.2	0.3	(17)	
	PrecStd	3.4	0.5	0.6	0.8	0.7	0.8	1.2	1.3	1.3	1.1	0.8	0.6	0.4	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.7	57.9	65.7	76.5	81.8	89.1	92.0	92.9	82.8	72.3	62.2	51.9	(19)	
		MCWB	48.1	48.1	52.5	60.2	65.4	69.8	71.1	69.0	66.7	60.5	54.8	47.0	(20)	
	2%	DB	49.2	53.3	61.2	71.4	77.6	84.5	87.7	87.5	77.9	66.5	57.4	47.9	(21)	
		MCWB	44.1	46.3	50.3	57.9	63.9	69.3	70.1	68.9	64.3	58.1	51.8	43.7	(22)	
	5%	DB	45.0	49.5	57.0	67.6	74.5	81.0	84.3	83.5	73.8	63.1	53.6	45.0	(23)	
		MCWB	41.0	43.7	47.8	55.5	62.1	66.8	68.0	67.5	62.3	56.4	50.0	41.6	(24)	
	10%	DB	41.6	45.5	53.3	64.2	71.1	77.7	80.8	80.1	70.3	60.1	50.5	42.1	(25)	
		MCWB	38.5	41.0	45.9	53.4	60.0	65.1	66.5	65.7	60.5	55.1	47.5	39.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.6	49.3	53.5	61.7	67.9	73.7	73.7	73.4	67.5	61.8	55.4	47.6	(27)	
		MCDB	54.6	56.1	63.4	74.3	78.0	84.6	86.8	88.7	79.6	69.6	60.8	51.3	(28)	
	2%	WB	44.5	46.7	50.8	58.5	65.2	70.8	71.2	70.3	65.4	59.4	53.0	44.3	(29)	
		MCDB	48.6	52.2	59.3	69.8	75.3	81.5	84.3	83.3	75.3	64.5	56.6	47.2	(30)	
	5%	WB	41.4	44.2	48.8	56.3	63.1	68.4	69.3	68.6	63.4	57.6	50.3	42.0	(31)	
		MCDB	44.5	49.2	55.7	66.3	72.5	78.2	81.2	81.1	71.6	61.8	53.2	44.5	(32)	
	10%	WB	38.6	41.3	46.6	54.3	61.1	66.3	67.5	66.9	61.6	55.7	47.8	39.9	(33)	
		MCDB	41.2	45.4	52.4	63.0	69.7	75.4	78.3	78.0	68.8	59.6	50.2	42.0	(34)	
(35) Mean Daily Temperature Range	MDBR		7.7	9.8	12.8	15.2	15.3	15.9	16.6	16.3	13.9	11.3	8.0	7.1	(35)	
	5% DB	MCDBR	12.1	14.8	18.9	19.3	19.7	20.6	21.4	21.2	19.1	15.7	12.3	11.4	(36)	
		MCWBR	8.8	9.5	10.7	9.7	9.5	9.1	8.1	8.2	8.5	8.8	8.0	8.6	(37)	
	5% WB	MCDBR	11.8	14.6	17.6	18.4	18.8	19.1	19.9	20.1	17.4	14.1	11.3	10.7	(38)	
		MCWBR	8.9	9.8	10.5	9.7	9.6	9.4	8.3	8.6	8.5	8.5	7.8	8.5	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.307	0.330	0.370	0.413	0.414	0.417	0.425	0.422	0.394	0.365	0.339	0.306	(40)	
	taud		2.374	2.347	2.282	2.196	2.234	2.259	2.239	2.262	2.311	2.365	2.391	2.409	(41)	
	Ebn at Noon		238	259	265	264	269	268	264	259	254	242	222	223	(42)	
	Edh at Noon		22	28	35	42	42	42	42	39	34	27	21	18	(43)	
(44) All-Sky Solar Radiation	RadAvg		300	570	917	1377	1613	1758	1687	1506	1073	646	325	235	(44)	
	RadStd		46	75	128	160	184	147	168	147	139	81	34	26	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+1.51	N/A	N/A	+2.23	N/A	N/A	-279	-405	+266	+110	(46)
(47)	Regional (88 neighbors)	+0.97	N/A	N/A	+1.40	+0.72	+0.61	-220	-288	+140	+61	(47)

Nomenclature: See separate page